

Flask Application Deployment Guide (Gunicorn + Nginx on Ubuntu VPS)

Overview

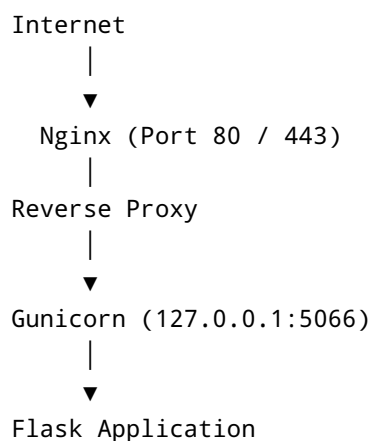
This guide explains how to deploy our Flask application to an Ubuntu VPS using:

- **Gunicorn** – Runs the Flask application.
- **Nginx** – Acts as a reverse proxy, accepts incoming HTTP/HTTPS requests, and forwards them to Gunicorn.
- **systemd** – Keeps the Gunicorn service running and starts it automatically after a reboot.

Current Application Information

Item	Value
Project Directory	/Documents/RoundAbout
Flask Application	/Documents/RoundAbout/app
WSGI Entry Point	/Documents/RoundAbout/app/wsgi.py
Static Files	/Documents/RoundAbout/app/static
Gunicorn Port	127.0.0.1:5066

Deployment Flow



Gunicorn should **only** listen on localhost (`127.0.0.1`) so it cannot be accessed directly from the internet. Nginx is the only public-facing service.

Step 1 – Install Required Packages

Update Ubuntu first:

```
sudo apt update  
sudo apt upgrade -y
```

Install Nginx:

```
sudo apt install nginx -y
```

Verify the installation:

```
nginx -v
```

Start the service:

```
sudo systemctl start nginx
```

Enable it to start automatically on boot:

```
sudo systemctl enable nginx
```

Check its status:

```
sudo systemctl status nginx
```

If everything is working, opening the VPS IP address in a browser should display the default "**Welcome to nginx!**" page.

Step 2 – Install Gunicorn

Activate the application's virtual environment:

```
source /Documents/RoundAbout/venv/bin/activate
```

Install Gunicorn:

```
pip install gunicorn
```

Test that Gunicorn can start the application:

```
cd /Documents/RoundAbout/app  
  
gunicorn --bind 127.0.0.1:5066 wsgi:app
```

Open another terminal and verify:

```
curl http://127.0.0.1:5066
```

If the application responds successfully, Gunicorn is configured correctly.

Stop Gunicorn with:

```
CTRL + C
```

Step 3 – Create the Gunicorn Service

Create the service file:

```
sudo nano /etc/systemd/system/roundabout.service
```

Paste the following configuration:

```
[Unit]  
Description=Gunicorn service for RoundAbout  
After=network.target  
  
[Service]  
User=www-data  
Group=www-data
```

```
WorkingDirectory=/Documents/RoundAbout/app

Environment="PATH=/Documents/RoundAbout/venv/bin"

ExecStart=/Documents/RoundAbout/venv/bin/gunicorn
    --bind 127.0.0.1:5066
    wsgi:app

Restart=always

[Install]
WantedBy=multi-user.target
```

Reload systemd:

```
sudo systemctl daemon-reload
```

Enable the service:

```
sudo systemctl enable roundabout
```

Start the service:

```
sudo systemctl start roundabout
```

Verify it is running:

```
sudo systemctl status roundabout
```

Step 4 – Configure Nginx

Create a new site configuration:

```
sudo nano /etc/nginx/sites-available/roundabout
```

Paste:

```
server {  
    listen 80;  
  
    server_name _;  
  
    location /static/ {  
        alias /Documents/RoundAbout/app/static/;  
    }  
  
    location / {  
        proxy_pass http://127.0.0.1:5066;  
  
        proxy_set_header Host $host;  
        proxy_set_header X-Real-IP $remote_addr;  
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;  
        proxy_set_header X-Forwarded-Proto $scheme;  
    }  
}
```

Enable the site:

```
sudo ln -s /etc/nginx/sites-available/roundabout /etc/nginx/sites-enabled/
```

(Optional) Remove the default site:

```
sudo rm /etc/nginx/sites-enabled/default
```

Test the configuration:

```
sudo nginx -t
```

Restart Nginx:

```
sudo systemctl restart nginx
```

Step 5 – Verify the Deployment

Check the following:

Gunicorn

```
sudo systemctl status roundabout
```

Nginx

```
sudo systemctl status nginx
```

Test Locally

```
curl http://127.0.0.1:5066
```

Test Through Nginx

```
curl http://localhost
```

or open:

```
http://<VPS-IP>
```

If everything is configured correctly, the Flask application should be accessible through the VPS IP address.

Useful Ubuntu Commands

Check service status

```
sudo systemctl status nginx  
sudo systemctl status roundabout
```

Restart a service

```
sudo systemctl restart nginx  
sudo systemctl restart roundabout
```

Stop a service

```
sudo systemctl stop nginx  
sudo systemctl stop roundabout
```

Start a service

```
sudo systemctl start nginx  
sudo systemctl start roundabout
```

Enable automatic startup

```
sudo systemctl enable nginx  
sudo systemctl enable roundabout
```

Disable automatic startup

```
sudo systemctl disable nginx  
sudo systemctl disable roundabout
```

Log Files

Gunicorn logs

```
sudo journalctl -u roundabout -f
```

Nginx access log

```
sudo tail -f /var/log/nginx/access.log
```

Nginx error log

```
sudo tail -f /var/log/nginx/error.log
```

Notes

- Gunicorn should only listen on `127.0.0.1:5066`.
- Nginx is the only service exposed to the internet.
- Static files are served directly by Nginx from `/Documents/RoundAbout/app/static`, improving performance and reducing load on the Flask application.
- All application management (start, stop, restart, boot) should be performed through the `roundabout` systemd service rather than running Gunicorn manually.
- Before exposing the application publicly, ensure the VPS firewall allows HTTP (port 80) and, once HTTPS is configured, HTTPS (port 443).